CAIE 9618 AS 计算机科学 · Unit 2 Communication

2.1 Networks including the internet — MS 原文背诵笔记

考纲	2027–2029 syllabus（version 2，2025 年 12 月）。2.1 的学习目标与 2024–2026 版相同，本笔记按其顺序排列。
检索范围（**9618**）	Paper 1，2021 May/June – 2026 May/June 的全部 May/June 与 Oct/Nov 场次（variant 11/12/13，共 33 份 MS）。2026 Oct/Nov 尚未公布。
检索范围（**9608**）	旧考纲 Paper 1，2015–2021 全部 May/June 与 Oct/Nov 场次（共 42 份 MS），只在 9618 MS 没有或太少的知识点使用。
未纳入	Feb/March 场次（9618 Paper 12 / 9608 Paper 12）在公开渠道（PapaCambridge、PMT 等）都下载不到，未能检索。

标记说明

MS 最全的一张原 MS 截图（含 Question / Answer / Marks 原表格）。

同类 **MS** 其他场次出现过的同类评分点，内容已包含在上图中，只列出处、不重复截图。

＋ 额外得分点 其他场次 MS 中有、而上图没有的独立得分点，直接截取原 MS 的对应行。

题目情景 **QP** 只在 MS 依赖题目情景（才能看懂）时附上题干截图。

9608 旧考纲 来自旧考纲 9608 的 MS，用于补充 9618 没有或太少的知识点。

⚠️ 未找到 **MS** 9618 与 9608 都没有 MS。若旧笔记有内容则保留并注明"非 MS"，下方留空白供补笔记。

注：2026 May/June 的 MS 来自 PapaCambridge，原 PDF 自带淡色网站水印；部分 9608 旧卷的右上角有网站角标。截图保持原样，未做修改。

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1 Purpose and benefits of networking devices | 网络设备的目的与好处

1.1 Purpose and benefits of networking devices

MS 9618 · 2023 May/June · Paper 12 · Q1(a)

Question	Answer	Marks
1(a)	**1 mark** each to **max 2** • Allows the sharing of files/data // Allows communication between the devices • Allows the sharing of resources e.g. hardware / software (applications) • Allows central management // by example, backup, security, etc.	**2**

2 Characteristics of a LAN and a WAN | LAN 与 WAN 的特征

2.1 Characteristics of a LAN

MS 9618 · 2025 Oct/Nov · Paper 12 · Q5(a)

Question	Answer	Marks
5(a)	**1 mark** per bullet point, **max 2** marks • Covers a small geographical area • Normally privately owned / dedicated infrastructure • Can use wired or wireless communication	**2**

同类 **MS**（已包含在上图中）：9618 · 2023 May/June · Paper 12 · Q1(b)

2.2 Characteristics of a WAN

MS 9618 · 2024 Oct/Nov · Paper 13 · Q9(a)

Question	Answer	Marks
9(a)	**1 mark** for each bullet point (**max 2**) • A WAN covers a large geographical area • External/public infrastructure is used • Non-dedicated hardware	**2**

2.3 Differences between a LAN and a WAN

MS 9618 · 2023 May/June · Paper 13 · Q2(a)

Question	Answer	Marks
2(a)	**1 mark** each to **max 2** • WAN covers a large **geographical** area and LAN covers a small **geographical** area • LAN connections between devices are usually physical, whereas the WAN connections are often virtual • A LAN has a high data transfer rate, whereas a WAN has a low data transfer rate • The **ownership** of a LAN is private; the **ownership** of a WAN can be private or public • LAN is **usually** more secure than a WAN because protection is easier to implement	**2**

+ 额外得分点 9618 · 2025 May/June · Paper 12 · Q6(a)

- A WAN may use non-dedicated hardware the hardware in a LAN is usually dedicated
- A WAN uses external / not company owned transmission media // the transmission media for a LAN is usually company owned

2.4 Identifying a LAN in a scenario

MS 9618 · 2021 Oct/Nov · Paper 11 · Q8(a)

Question	Answer	Marks
8(a)	**1 mark** per bullet point • LAN • Small geographical area • No leasing external infrastructure / transmission media // does not use internet to transmit within the building	**3**

同类 **MS**（已包含在上图中）：9618 · 2021 Oct/Nov · Paper 13 · Q8(a)

3 Client-server and peer-to-peer models | 客户端-服务器模型与对等网络模型

3.1 Client-server: what it is / roles of the computers

题目情景 **QP** 9618 · 2025 May/June · Paper 13 · Q3(c)

3 A programmer is writing a program in a high-level language.

(c) The computer program allows users to play a game in a virtual world. Users can play the game from any computer with internet access using a web browser. The game allows users to interact with other users.

Explain the reasons why the statement 'This computer game uses a client-server model' is correct.

MS 9618 · 2025 May/June · Paper 13 · Q3(c)

Question	Answer	Marks
3(c)	**1 mark** each to **max 4** • A webserver stores all the data for each player • Each player is on a client computer // The player's web browser is the client • that sends requests over the internet to the web server • the server performs the required action in the game • the server updates the data in the game • the server sends the results to the player	**4**

+ 额外得分点 9618 · 2024 May/June · Paper 11 · Q5(a)

- Description e.g. Receives requests, processes the requests

+ 额外得分点 9618 · 2024 May/June · Paper 11 · Q5(a)

- Description e.g. Sends request to the server, waits and outputs the response

MS **9608** 旧考纲 9608 · 2015 Oct/Nov · Paper 11 · Q7(a)(i)

7 (a) (i)
– at least one computer used to "serve" …
– … other computers are referred to as "clients"
– server provides services / applications etc. …
– … which may be requested by clients [2]

同类 **MS**（已包含在上图中）：9608 · 2019 Oct/Nov · Paper 11 · Q4(a)(i)；9608 · 2020 May/June · Paper 13 · Q1(b)；9608 · 2021 Oct/Nov · Paper 11 · Q11(a)

3.2 Client-server: benefits

MS **9608** 旧考纲 9608 · 2015 Oct/Nov · Paper 11 · Q7(a)(ii)

(ii) any **two** from:

– files and resources are centralised
– creation of security / manage security
– user needs user name and password to access network
– centralised back-up
– intranet capability
– Internet monitoring
– clients can be less powerful machines, therefore less expensive to buy
– saving resources on server reduces the burden on the client [2]

同类 **MS**（已包含在上图中）：9608 · 2015 Oct/Nov · Paper 13 · Q7(a)(ii)

3.3 Client-server: drawbacks

⚠️ 未找到 **MS** 9618（2021–2026，33 份）和 9608（2015–2021，42 份）Paper 1 的 MS 里，都没有针对这一点的评分点。

原笔记内容（不是 **MS** 原文，来源可能是课本或自己总结，仅供参考）：

- *Initial set-up costs for the server and the client workstations will be high*
- *A server-based network requires the use of a specialist network operating system*
- *If the server fails then the work of all users is affected*

我的笔记：

3.4 Peer-to-peer: key features

MS 9618 · 2021 May/June · Paper 11 · Q4(a)

Question	Answer	Marks
4(a)	**1 mark** per bullet point to **max 2** • All computers are of equal status • Each computer provides access to resources and data // data is distributed • Computers can communicate and share resources • Each computer is responsible for its own security	**2**

同类 **MS**（已包含在上图中）：9618 · 2021 May/June · Paper 13 · Q4(a)

3.5 Peer-to-peer: benefits

⚠️ 未找到 **MS** 9618（2021–2026，33 份）和 9608（2015–2021，42 份）Paper 1 的 MS 里，都没有针对这一点的评分点。

原笔记内容（不是 **MS** 原文，来源可能是课本或自己总结，仅供参考）：

- *No dedicate additional computers*
- *additional computer can join the P2P network easily*
- *server is not required*
- *can act as both a provider and receiver of resources*

我的笔记：

3.6 Peer-to-peer: drawbacks

MS 9618 · 2021 May/June · Paper 11 · Q4(b)

Question	Answer	Marks
4(b)	**1 mark** per bullet point to **max 2** per drawback • Reduced security // no central management of security • … only as secure as the weakest computer on the network • … each computer is at risk from viruses from other computers • No central management of backup • ... if the data from one computer is not backed up it is lost to all of them • No central management of files/software • … consistency may be difficult to maintain • … each computer may have different software from the others • Individual computers may respond slower • … because they are being accessed by other computers • In order to share files etc. all the computers involved need to be switched on • ... so the files etc. may not be always available	**4**

同类 **MS**（已包含在上图中）：9618 · 2021 May/June · Paper 13 · Q4(b)

3.7 Justify the use of a model for a given situation

⚠️ 未找到 **MS** 9618（2021–2026，33 份）和 9608（2015–2021，42 份）Paper 1 的 MS 里，都没有针对这一点的评分点。

我的笔记：

4 Thin-client and thick-client | 瘦客户端与胖客户端

4.1 Thin-client

MS 9618 · 2026 May/June · Paper 11 · Q7(a)

Question	Answer	Marks	Guidance
7(a)	**1 mark** for each point to **max 2** • Data is not stored on the client computer // all data is stored on the server • The client computer is reliant on access to server • The client computer is heavily reliant on the network connection • The client computer requires few local resources, for example, memory / processing power / storage • The client computer performs **minimal functions** / processes	**2**	

同类 **MS**（已包含在上图中）：9618 · 2024 May/June · Paper 12 · Q3(b)

+ 额外得分点 9618 · 2023 May/June · Paper 13 · Q2(c)

- Server performs **all** processes required by the task and/or data storage
- Clients **only** sends requests to the server and displays the returned results

4.2 Thick-client

MS 9618 · 2023 May/June · Paper 12 · Q1(e)

Question	Answer	Marks
1(e)	**1 mark** each • The server performs **minimal** / **some processing** for the client • The clients also do **most** of their own processing/work independently // most of the resources are installed locally	**2**

4.3 Differences between thin-client and thick-client

没有一道 MS 直接问"两者区别"。可以对照 4.1 和 4.2 的 MS 自己总结。

⚠️ 未找到 **MS** 9618（2021–2026，33 份）和 9608（2015–2021，42 份）Paper 1 的 MS 里，都没有针对这一点的评分点。

我的笔记：

5 Bus, star, mesh and hybrid topologies | 总线、星型、网状与混合拓扑

5.1 Comparing bus, star and mesh

MS 9618 · 2024 May/June · Paper 11 · Q8(a)

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>8(a)</td><td>1 mark for each correct row:

<table>
<tr><th>Statement</th><th>Bus</th><th>Star</th><th>Mesh</th></tr>
<tr><td>all devices connect to one central device</td><td></td><td>✓</td><td></td></tr>
<tr><td>all devices connect to a central cable</td><td>✓</td><td></td><td></td></tr>
<tr><td>multiple paths for the packets to travel</td><td></td><td></td><td>✓</td></tr>
<tr><td>robust against damage because if any line fails, the rest of the network retains full functionality</td><td></td><td>✓</td><td>✓</td></tr>
<tr><td>most likely to lose data through collisions</td><td>✓</td><td></td><td></td></tr>
</table>
</td><td>5</td></tr>
</table>

5.2 Bus: how packets are transmitted between two hosts

MS 9618 · 2026 May/June · Paper 11 · Q7(b)(i)

Question	Answer	Marks	Guidance
7(b)(i)	**1 mark** for each point to **max 5** Bus: • Both nodes are connected to a bus/cable • Data is broadcast on the cable • but is only accepted by the intended recipient Ethernet: • Uses the IP/MAC addresses of the two nodes • Ethernet transmits data as frames • Ethernet uses Carrier Sense Multi-Access / Collision Detection / CSMA/CD • The transmitting node listens to see if the line is free **and** only submits data when the line is free • The node waits a random time if a collision is detected **and** then tries to send again	5	Award a maximum of 4 marks if only Ethernet referenced

说明：这道题同时考了 Bus 与 Ethernet，第 10.1 节也会用到。

5.3 Bus: drawbacks

MS 9618 · 2026 May/June · Paper 11 · Q7(b)(ii)

Question	Answer	Marks	Guidance
7(b)(ii)	**1 mark** for each point to **max 2** • Bus networks are inefficient with a large number of nodes • because only one transmission can be on the line at once • Bus networks are inefficient with large numbers of simultaneous data transmissions • because of the need for nodes to wait for the line to be free • Bus networks are not easily scalable • If the bus/cable goes down the whole/part of the network is unusable • Transmissions are broadcast to all nodes, so could be privacy/confidentiality/security issues	2	The answer spaces are numbered, so mark the first answer in each space

5.4 Bus: benefits

⚠️ 未找到 **MS** 9618（2021–2026，33 份）和 9608（2015–2021，42 份）Paper 1 的 MS 里，都没有针对这一点的评分点。

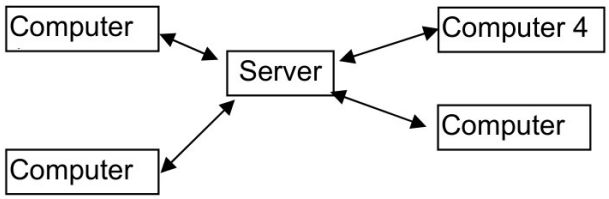
原笔记内容（不是 **MS** 原文，来源可能是课本或自己总结，仅供参考）：

- *Easier to set-up/extend*
- *Less cable required*

我的笔记：

5.5 Star: structure (drawing a star topology)

MS 9618 · 2023 May/June · Paper 12 · Q1(c)

Question	Answer	Marks
1(c)	**1 mark** each to **max 2** • Each computer directly connected only to the server ... • ... all components correctly labelled 	**2**

同类 **MS**（已包含在上图中）：9618 · 2022 Oct/Nov · Paper 12 · Q10(a)；9618 · 2023 Oct/Nov · Paper 12 · Q7(b)(i)；9618 · 2025 Oct/Nov · Paper 12 · Q5(b)

MS 9618 · 2021 Oct/Nov · Paper 12 · Q3(b)(i)

Question	Answer	Marks
3(b)(i)	**1 mark** for identification of star topology **1 mark** for justification Devices are connected directly to the router independently // all devices are **only** connected to the router	**2**

5.6 Star: how packets are transmitted between two hosts

MS 9618 · 2023 Oct/Nov · Paper 12 · Q7(b)(ii)

Question	Answer	Marks
7(b)(ii)	**1 mark** for each bullet point (**max 2**) • The data from the sending laptop is transmitted to the router • The data has address of recipient • The router determines recipient's destination address • ... by using a routing table • The router transmits data directly / only to recipient	**2**

同类 **MS**（已包含在上图中）：9618 · 2024 May/June · Paper 13 · Q5(b)；9618 · 2025 Oct/Nov · Paper 12 · Q5(c)

5.7 Star: benefits (justify star instead of bus)

MS 9618 · 2023 Oct/Nov · Paper 11 · Q2(d)

Question	Answer	Marks
2(d)	**1 mark** for each advantage and **1 mark** for valid corresponding expansion • Star topology is more resilient to faults • …because there is no single cable and leads to less disruption to teaching • Higher performance as fewer collisions • ...because each device in the classroom is only connected to the switch • Easier to add new nodes • …because each device in the classroom connects directly to the switch • Easier to fault find compared to bus topology	**2**

5.8 Mesh: what it is / how packets are transmitted

MS 9618 · 2023 May/June · Paper 13 · Q2(b)(i)

Question	Answer	Marks
2(b)(i)	**1 mark** each to **max 2** • All computers are connected to at least one other device • There are multiple routes between devices • The computers can act as relays, passing packets on towards the final destination	**2**

5.9 Mesh: benefits (instead of bus)

MS 9618 · 2023 May/June · Paper 13 · Q2(b)(ii)

Question	Answer	Marks
2(b)(ii)	**1 mark** each to **max 2** • If one line goes down there are more routes available • Improved security as not using one main line • No/fewer collisions • New nodes can be added without interruption or interfering with other nodes • More secure because data is sent over a dedicated connection	**2**

5.10 Hybrid topology

⚠️ 未找到 **MS** 9618（2021–2026，33 份）和 9608（2015–2021，42 份）Paper 1 的 MS 里，都没有针对这一点的评分点。

原笔记内容（不是 **MS** 原文，来源可能是课本或自己总结，仅供参考）：

- *Definition: a collection of connected LANs where some of them have different topologies or supporting technologies*
- *Benefits: It is of an advantage to be able to connect a new topology LAN to existing LANs where it is not sensible or not possible to use the existing topology for the new LAN. An example is when a wired LAN is already installed but a new wireless LAN is to be connected to it.*

我的笔记：

6 Cloud computing | 云计算

6.1 What is cloud computing

MS 9618 · 2022 May/June · Paper 13 · Q8(a)

Question	Answer	Marks
8(a)	Accessing a service/files/software on a remote **server**	**1**

6.2 Public cloud and private cloud

MS 9618 · 2022 May/June · Paper 13 · Q8(b)

Question	Answer	Marks
8(b)	**1 mark** each from: Public e.g. • Computing services offered by 3rd party provider over the public Internet • Public is open/available to anyone with the appropriate equipment/software/credentials Private e.g. • Computing services offered either over the Internet or a private internal network • Only available to select users not the general public • Private is a dedicated/bespoke system only accessible for/from the organisation	**2**

同类 **MS**（已包含在上图中）：9618 · 2024 May/June · Paper 13 · Q5(a)(i)；9618 · 2026 May/June · Paper 13 · Q3(c)(ii)；9618 · 2022 Oct/Nov · Paper 13 · Q7(a)

6.3 Why use a public cloud (scenario)

题目情景 **QP** 9618 · 2023 Oct/Nov · Paper 13 · Q3(a)

3 A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

(a) (i) Explain why the company uses a public cloud to store these courses.

MS 9618 · 2023 Oct/Nov · Paper 13 · Q3(a)(i)

Question	Answer	Marks
3(a)(i)	**1 mark** for each bullet point (**max 2**). • Courses must be available to anyone who wishes to follow them • Courses must be available on the internet • Company is willing to share infrastructure with other companies (public) • …which is more economic for the company	**2**

6.4 Benefits of a private cloud instead of a public cloud

MS 9618 · 2024 May/June · Paper 13 · Q5(a)(ii)

Question	Answer	Marks
5(a)(ii)	**1 mark** each to **max 3**: e.g. • Not reliant on a third party • … gives greater **control** over security/privacy • … gives greater **control** over backup • Storage can be tailored/scalable to company requirements// an example e.g. the amount of storage accessible/ facilitating the sharing of files	**3**

6.5 Benefits of cloud computing / cloud storage

MS 9618 · 2021 May/June · Paper 12 · Q5(c)(i)

Question	Answer	Marks
5(c)(i)	**1 mark** per bullet point to **max 2** • Cloud storage can be free (for small quantities) • No need for separate (high capacity) storage devices // saves storage on existing devices • Can access data from any computer **with internet access** • Most cloud data services will have in-built backup/disaster recovery • Security could be better • Can easily increase capacity • Data can be easily shared	**2**

同类 **MS**（已包含在上图中）：9618 · 2025 Oct/Nov · Paper 12 · Q8(a)(i)；9618 · 2022 May/June · Paper 13 · Q8(c)；9618 · 2026 May/June · Paper 13 · Q3(c)(iii)

+ 额外得分点 9618 · 2022 May/June · Paper 13 · Q8(c)

- Do not need to buy specific software/hardware

+ 额外得分点 9618 · 2022 May/June · Paper 13 · Q8(c)

- Can have multiple people working on the same document

+ 额外得分点 9618 · 2026 May/June · Paper 12 · Q6(d)

- The user can upgrade/change their device and retain old data // the data can be accessed from different appropriate devices

6.6 Drawbacks of cloud computing / cloud storage

MS 9618 · 2025 Oct/Nov · Paper 12 · Q8(a)(ii)

Question	Answer	Marks
8(a)(ii)	**1 mark** per bullet point, **max 2** marks • The storage/cloud can only be accessed with an internet connection • There may be no backups // possibly no control over backups // reliance on third party for backups • Uploading / downloading data could take a long time • Potential issues with the cloud storage company impacting access e.g. downtime or technical problems • Potential compatibility issues if wanting to move the data • Limited storage availability without paying extra • … meaning (possibly) more expensive in the long term • Possibly security issues	**2**

同类 **MS**（已包含在上图中）：9618 · 2021 May/June · Paper 12 · Q5(c)(ii)；9618 · 2022 May/June · Paper 13 · Q8(c)；9618 · 2023 Oct/Nov · Paper 11 · Q2(c)

+ 额外得分点 9618 · 2026 May/June · Paper 12 · Q6(d)

- The data could be sold or misused by the cloud owner

6.7 Disadvantages of a public cloud compared with a server on a LAN

题目情景 **QP** 9618 · 2023 Oct/Nov · Paper 13 · Q3(a)(ii)

(ii) Describe **two** disadvantages of storing data on a public cloud compared to storing data on a server in a Local Area Network (LAN).

MS 9618 · 2023 Oct/Nov · Paper 13 · Q3(a)(ii)

Question	Answer	Marks
3(a)(ii)	**1 mark** for each bullet point (**max 2** for each disadvantage). • There could be a possible loss of control unlike the LAN • …because the data is stored on a remote infrastructure / someone else's infrastructure • …reliance on external agency to complete tasks, e.g. backups, security • Requires reliable internet connection • …to ensure access to the remote data, more likely with LAN • Increased recurring costs • ...as cloud provider charges must be paid, costs for LAN once only.	**4**

7 Wired and wireless networks | 有线与无线网络

7.1 Characteristics of the media (fibre-optic, radio waves / microwaves)

MS 9618 · 2024 Oct/Nov · Paper 13 · Q9(b)

Question	Answer	Marks
9(b)	**1 mark** for correct media **1 mark** for matching description e.g. • Fibre optic • Transmits data as pulses of light • Radio waves / microwaves • Transmits electromagnetic waves on different frequencies	**4**

7.2 Characteristics of the media (copper cable, satellite, radio waves)

9618 MS 里没有描述 copper cable 和 satellite 的题，下面用 9608 旧考纲补充。

MS 9608 旧考纲 9608 · 2020 May/June · Paper 13 · Q1(a)

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>1(a)</td><td>1 mark for each correctly completed media or description

<table>
<tr><th>Type of connection</th><th>Description</th></tr>
<tr><td>Fibre-optic</td><td>Transmits data as light //
Uses (a bundle of) glass/plastic threads to transmit data</td></tr>
<tr><td>Satellite</td><td>A communication device in Earth’s orbit that receives and transmits data</td></tr>
<tr><td>Radio Waves</td><td>Carries data wirelessly, often known as Wi-Fi //
Carries data in the form of electromagnetic waves</td></tr>
<tr><td>Copper cable</td><td>Carries data as electrical signals and can consist of a twisted pair</td></tr>
</table></td><td>4</td></tr>
</table>

7.3 Differences between fibre-optic and copper cable

MS 9618 · 2022 May/June · Paper 12 · Q9(b)

Question	Answer	Marks
9(b)	**1 mark** for each difference e.g. • Fibre optic data is transmitted using light, copper cable through electrical signals • Fibre optic has higher bandwidth than copper cable // Fibre optic has higher transmission rates than copper cable • Fibre optic has smaller risk of (noise) interference than copper cable • Fibre optic can be used over longer distances than copper cable before repeaters are needed • Fibre optic is much more difficult to hack into than copper cable • Fibre optic is more prone to damage than copper cable	**3**

7.4 Benefits of fibre-optic vs copper cable (including benefits of copper)

MS 9608 旧考纲 9608 · 2015 Oct/Nov · Paper 11 · Q6

6 any **four** points from (maximum 3 marks per type of cable):

– fibre optic cables have greater bandwidth
– fibre optic cables need less signal boosting // can transmit over longer distances
– fibre optic cables have greater security (more difficult to “tap” into)
– fibre optic cables are immune to electromagnetic and other effects
– fibre optic cabling is lighter in weight (easier to install)
– fibre optic cables consume less power
– copper cabling is less expensive to install
– copper cable is easier to install because it is more flexible
– it is easier to make terminations using copper cabling
– the expertise in use of copper cabling is more extensive
– has been around for years … so very little is “unknown” about installations using this type of cabling [4]

同类 **MS**（已包含在上图中）：9608 · 2015 Oct/Nov · Paper 13 · Q6；9608 · 2018 Oct/Nov · Paper 12 · Q2(c)(i)；9608 · 2019 Oct/Nov · Paper 11 · Q4(c)(i)

7.5 Drawbacks of fibre-optic cable

MS **9608** 旧考纲 9608 · 2018 Oct/Nov · Paper 12 · Q2(c)(ii)

Question	Answer	Marks
2(c)(ii)	**1 mark** per bullet point to **max 2** • (Initial) installation cost is higher // Cable / hardware is more expensive to buy (per metre) • Specialists / trained personnel needed to install / maintain • Difficult to terminate // Electronics at both ends are more complex • Fibre-optic cables can break when bent • Only transmits data in one direction • If a fibre-optic cable connection fails, many more services can be affected	**2**

\+ 额外得分点 **9608** 旧考纲 9608 · 2019 Oct/Nov · Paper 11 · Q4(c)(ii)

- Only transmits data in one direction // Cannot transmit power, only data

7.6 Satellites instead of copper cables

MS 9618 · 2022 Oct/Nov · Paper 13 · Q7(b)

Question	Answer	Marks
7(b)	**1 mark** for a correct advantage: • not fixed to a single location • allows access in remote / rural areas **1 mark** for each correct disadvantage (**max 2**): • high latency / lag / slow to connect • more expensive than wired methods, as need extra equipment • signal is affected by **bad weather** • the transmission speed is slower than fixed line broadband • direct line of sight needed	**3**

7.7 Wired vs wireless: benefits of each

题目情景 **QP** 9618 · 2021 May/June · Paper 11 · Q4(c)(ii)

(c) Melinda connects her laptop to the internet through her router.

(ii) Melinda mainly uses the internet to watch films and play computer games.

Tick (✓) **one** box to identify whether Melinda should connect to the router using a wired or wireless network **and** justify your choice.

Wired	
Wireless	

MS 9618 · 2021 May/June · Paper 11 · Q4(c)(ii)

Question	Answer	Marks
4(c)(ii)	**1 mark** per bullet point for justification up to **max 3** No mark for identification of wired/wireless Wired • Faster connection // higher bandwidth • …. needed as she is downloading/streaming large files • … less time waiting / less latency / fewer delays • **More** reliable / stable connection • … is less susceptible to issues with distance/walls/interference • **More** secure Wireless • Freedom of movement • … can move between different rooms with a mobile device and still receive/transmit data • … no need of a physical connection • Easily expanded if friends want to access the same network • Less cabling / expertise is needed • ... making the initial setup less expensive	**3**

同类 **MS**（已包含在上图中）：9618 · 2021 May/June · Paper 13 · Q4(c)(ii)；9618 · 2025 Oct/Nov · Paper 13 · Q4(a)；9608 · 2018 Oct/Nov · Paper 13 · Q7(a)

\+ 额外得分点 9618 · 2025 Oct/Nov · Paper 13 · Q4(a)

- Multiple different device types can be connected

7.8 Wireless: drawbacks

MS 9618 · 2025 Oct/Nov · Paper 13 · Q4(b)

Question	Answer	Marks
4(b)	**1 mark** for a drawback • Can be less secure • **Transmission speed** can be slow • Interference is possible • The signal degrades quickly without repeaters/boosters	**1**

同类 **MS**（已包含在上图中）：9608 · 2018 Oct/Nov · Paper 13 · Q7(b)

7.9 Why provide both wired and wireless connections (scenario)

题目情景 **QP** 9618 · 2023 May/June · Paper 13 · Q2(d)

2 A university has two sites. Each site has several computer rooms. The computers are all connected as a WAN (wide area network).

(d) Students can connect their devices to the university network using cables or a wireless connection.

Explain the benefits to the students of allowing them to use **both** wired and wireless connections.

MS 9618 · 2023 May/June · Paper 13 · Q2(d)

Question	Answer	Marks
2(d)	**1 mark** each to **max 4** Examples: • Some students might only have one sort of connection on their device • Wired provides better performance for the student's device • ... for example, enabling faster access to university databases • There will be less interference if students connect via a cable • Students can transmit private/confidential data/work securely • ... for example, their final dissertation • Wireless connection means that the students can use their devices in different rooms/sites/outside/anywhere more freely // student devices can be portable • Wireless connection enables the students to bring multiple devices // bring their own devices // change devices	**4**

8 Hardware used to support a LAN | 支持 LAN 的硬件

8.1 Router / WNIC / Repeater / WAP (descriptions)

MS 9618 · 2022 May/June · Paper 12 · Q9(a)

Question	Answer	Marks
9(a)	**1 mark** for each completed name or description	**4**

Device	Description
Router	Receives and sends data between two networks operating on the same protocol
Wireless Network Interface Card (WNIC)	Hardware component that allows a device to connect to a wireless network // Provides a MAC address to the device to identify it on the wireless network
Repeater	Restores the digital signal so it can be transmitted over greater distances
Wireless Access Point (WAP)	Hardware component that provides radio communication from the central device to nodes on the network (and vice versa)

8.2 Switch / WAP / Bridge (purpose)

MS 9618 · 2023 Oct/Nov · Paper 11 · Q2(b)

Question	Answer	Marks
2(b)	Switch: **1 mark** for each bullet point (**max 1**) • To allow two or more devices to communicate with one another • To connect individual devices to each other • To receive transmissions and **forward** them to their destination Wireless Access Point (WAP): **1 mark** for each bullet point (**max 1**) • To allow connection of devices (to the central device) **using radio signals** / Wi-Fi • To allow the central device to send / receive **radio signals** / Wi-Fi signals • To allow wireless enabled devices to connect to a wired network Bridge:**1 mark** for each bullet point (**max 1**) • To connect two LANs / segments with the **same protocol** • To transmit data between two networks with the **same protocol**	**3**

8.3 Switch: functions

MS 9618 · 2025 May/June · Paper 12 · Q6(e)

Question	Answer	Marks
6(e)	**1 mark** each to **max 3** e.g. • To store the MAC addresses of devices connected to it • To receive packets from devices • To forward packets directly to the intended recipient • To provide a central point of connection for the network • to enable connected devices to communicate	**3**

8.4 Server

9618 MS 里没有专门定义 server 的题，下面用 9608 旧考纲补充。

MS 9608 旧考纲 9608 · 2021 May/June · Paper 11 · Q9(b)

Question	Answer	Marks
9(b)	**1 mark** for each correct term **Description** / **Term**: Receives data packets from a network and forwards them onto a similar network — **Router** Manages access to a centralised resource — **Server** Joins networks that use different sets of rules to transmit data — **Gateway** Monitors and controls incoming and outgoing network traffic based on set criteria — **Firewall**	**4**

MS 9608 旧考纲 9608 · 2019 May/June · Paper 12 · Q1(d)

Question	Answer	Marks
1(d)	**1 mark** per bullet point to **max 3** for any valid answer For example: • File server • Print server • Proxy server • Web server • Application server	**3**

同类 **MS**（已包含在上图中）：9608 · 2018 May/June · Paper 11 · Q5(a)

8.5 NIC (Network Interface Card)

⚠️ 未找到 **MS** 9618（2021–2026，33 份）和 9608（2015–2021，42 份）Paper 1 的 MS 里，都没有针对这一点的评分点。

原笔记内容（不是 **MS** 原文，来源可能是课本或自己总结，仅供参考）：

- *NIC: Hardware component that allows a device to connect to a wired network*

我的笔记：

8.6 WNIC: functions

MS 9618 · 2021 Oct/Nov · Paper 11 · Q8(c)

Question	Answer	Marks
8(c)	**1 mark** per bullet point to **max 4** • Provide interface to wireless network • … as an antenna • Receives analogue radio waves • … convert them to digital / binary • Checks incoming transmissions for correct MAC / IP address • … ignore transmissions not intended for it • Encrypts / encodes the data • Decrypts / decodes the data • Takes digital/binary input and converts to analogue waves • … sends the radio waves via the antenna	**4**

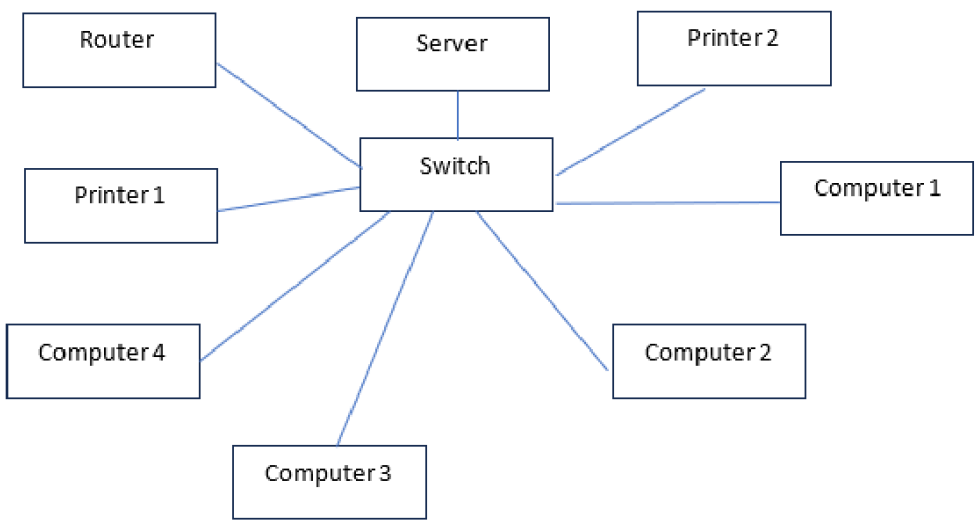
同类 **MS**（已包含在上图中）：9618 · 2021 Oct/Nov · Paper 13 · Q8(c)

8.7 WAP, cables, bridge, repeater

WAP 见 8.1 与 8.2；bridge 见 8.2；repeater 见 8.1；cables（copper / fibre-optic）见第 7 节。

8.8 Designing a LAN (diagram)

MS 9618 · 2025 May/June · Paper 12 · Q6(d)

Question	Answer	Marks
6(d)	**1 mark** each • Server directly connected to the switch • Four labelled computers all connected directly to the switch • Two printers each directly connected to the switch / server / a computer • Router / modem directly connected to the switch / server for internet access e.g. Example only  	**4**

9 Role and function of a router | 路由器的作用与功能

9.1 Functions of a router

MS 9618 · 2022 Oct/Nov · Paper 12 · Q10(b)(ii)

Question	Answer	Marks
10(b)(ii)	**1 mark** for each bullet point **(max 3)**: • receive packets from devices / internet • find destination of packets using the IP address • forward packets to **the destination** • assign **private** IP addresses to devices on LAN • store/update/maintain a routing table • find most efficient **path** to destination • maintain table of MAC and IP addresses • provides the LAN with a public IP address • acts as a gateway • performs protocol conversion • acts as a firewall	**3**

同类 **MS**（已包含在上图中）：9618 · 2021 Oct/Nov · Paper 12 · Q3(b)(ii)；9618 · 2023 Oct/Nov · Paper 11 · Q2(a)；9618 · 2024 May/June · Paper 12 · Q3(c)(i)；9618 · 2021 May/June · Paper 11 · Q4(c)(i)

+ 额外得分点 9618 · 2023 Oct/Nov · Paper 11 · Q2(a)

- Changes the packet format for transmission over the next network //
- Network Address Translation (NAT):NAT is a technique used by routers to allow multiple devices in a private local area network (LAN) to share a single public IP address.

10 Ethernet and CSMA/CD | 以太网与 CSMA/CD

10.1 Ethernet

MS 9618 · 2023 May/June · Paper 12 · Q1(d)

Question	Answer	Marks
1(d)	**1 mark** each to **max 3** • A protocol (suite) • For **data transmission** over standard / universal **wired** / **cabled** network connections • Uses Carrier Sense Multiple Access / Collision Detection (CSMA/CD) • Data is transmitted in frames • … each frame has a source and destination (IP/MAC) address • ... and error checking data (so damaged frames can be resent)	**3**

同类 **MS**（已包含在上图中）：9618 · 2026 May/June · Paper 11 · Q7(b)(i)（见 5.2）

10.2 How CSMA/CD detects and avoids collisions

MS 9618 · 2022 Oct/Nov · Paper 11 · Q8

Question	Answer	Marks
8	**1 mark** for bullet point (**max 4**): • CSMA/CD is a protocol used to detect and prevent collisions in a bus topology • before transmitting, a **device** checks if the channel is busy • If the channel is busy the device waits // if the channel is free the data is sent • because there is more than one computer connected to the same transmission medium • ... two workstations can start to transmit at the same time, causing a collision • If a collision is detected by the device, transmission is aborted / a jamming signal is transmitted • both devices wait a (different) **random** time and then try again	**4**

同类 **MS**（已包含在上图中）：9618 · 2022 May/June · Paper 12 · Q9(c)；9618 · 2023 Oct/Nov · Paper 12 · Q7(c)；9618 · 2024 May/June · Paper 13 · Q5(c)(i)；9618 · 2025 Oct/Nov · Paper 12 · Q5(d)

+ 额外得分点 9618 · 2025 Oct/Nov · Paper 12 · Q5(d)

- The random time is increased if there are multiple collisions

10.3 Drawbacks of CSMA/CD

MS 9618 · 2024 May/June · Paper 13 · Q5(c)(ii)

Question	Answer	Marks
5(c)(ii)	**1 mark** each to **max 2**: • Random time increased each time so can be infinite waiting • May be constant jamming signal so nothing ever sends • Certain nodes cannot be prioritised • High power consumption • Only suitable for short distance network // limited distance • Not scalable // more nodes means exponentially longer waiting times	**2**

11 Bit streaming | 比特流

11.1 What is bit streaming

MS 9618 · 2024 May/June · Paper 11 · Q2(e)(i)

Question	Answer	Marks
2(e)(i)	**1 mark** for: Continuous ordered flow of bits over a communication path	**1**

MS **9608** 旧考纲 9608 · 2015 Oct/Nov · Paper 11 · Q1(a)

1 **(a)** any **two** from:

– sequence of digital signals / bits
– over a communication path / Internet
– transfer of data at high speed
– requires fast broadband connection
– requires some form of buffering
– bits arrive in the same order as sent [2]

同类 **MS**（已包含在上图中）：9608 · 2015 Oct/Nov · Paper 13 · Q1(a)

11.2 Real-time vs on-demand: definitions

MS **9608** 旧考纲 9608 · 2016 May/June · Paper 13 · Q6(c)

(c) **One pair** from: **[2]**

- Real-time - a live stream of an event that is currently taking place
- On-demand - streaming of an event/programme that has taken place in the past

- Real time – the event is captured live with a video camera connected to a computer
- On-demand – Existing media are encoded to bit streaming format and uploaded to a server

- Real-time – cannot be paused / rewound etc
- On-demand – can be paused / re-wound / fast forwarded etc

11.3 Real-time vs on-demand: differences

MS 9618 · 2024 May/June · Paper 11 · Q2(e)(ii)

Question	Answer	Marks
2(e)(ii)	**1 mark** each to **max 2**: • Real-time is direct from source whereas on-demand is pre-recorded/downloaded to view later • Real-time cannot be re-watched, on-demand can be paused, re-watched etc. • Real-time plays continually, on-demand downloads sections/blocks and cannot play until next section is downloaded	**2**

11.4 How real-time bit streaming works

MS 9618 · 2025 May/June · Paper 11 · Q2(a)

Question	Answer	Marks
2(a)	**1 mark** each to **max 4** • Video is transmitted continuously • as a series of bits • The video is uploaded to a media server • The users download from the media server • On download, the media server sends the data to a buffer on the user's device • buffer is used when there is a difference in speed between transmission and receipt • Buffer stores data from server until recipient can receive it • Recipient views bit stream from the buffer	**4**

同类 **MS**（已包含在上图中）：9618 · 2024 Oct/Nov · Paper 13 · Q9(c)；9608 · 2019 Oct/Nov · Paper 12 · Q6(e)(i)

+ 额外得分点 9618 · 2024 Oct/Nov · Paper 13 · Q9(c)

- Data is compressed before transmitting

11.5 How on-demand bit streaming works (and real-time)

9618 MS 里没有讲 on-demand 流程的题，下面用 9608 旧考纲补充。

MS **9608 旧考纲** 9608 · 2015 Oct/Nov · Paper 11 · Q1(c)

(c) 2 marks for on-demand and 2 marks for real-time

on-demand

– digital video tape, analogue video tape, or digital files are converted to bit streaming – format for broadcasting on the net; this is known as encoding, these encoded streaming video files are then uploaded to a dedicated server
– a link for the encoded video is placed on a web site
– a user clicks on the link to download the encoded streaming video; the streamed video is
– then broadcast to the user as and when they require it
– can be paused / can go back and re-watch / fast-forward, etc.

real-time

– an event is captured live with a video camera
– the video camera is connected to a computer
– the video signal is converted to streaming media files (encoded) on the computer
– the encoded feed is then uploaded from the computer to a dedicated streaming server via cable, DSL, or a high-speed internet connection
– the server then sends the live images it to all users requesting it as real-time video streaming
– cannot be paused etc. [4]

同类 **MS**（已包含在上图中）：9608 · 2015 Oct/Nov · Paper 13 · Q1(c)

11.6 Benefits and problems of bit streaming

MS **9608** 旧考纲 9608 · 2015 Oct/Nov · Paper 11 · Q1(b)(i)

(b) **(i)** any **two** from:

– no need to wait for a whole file to be downloaded
– no need to store large files on user's computer
– allows on demand playback
– no specialist software is required for playback in browser [2]

MS **9608** 旧考纲 9608 · 2015 Oct/Nov · Paper 11 · Q1(b)(ii)

(ii) any **two** from:

– video stops / hangs if very slow Internet / broadband speed low
– video stops / hangs if inadequate buffering capacity
– loss of Internet means can't access films / files
– may require specific software to run the files / films
– viruses can be downloaded from the websites [2]

11.7 Importance of bit rate / broadband speed

题目情景 **QP** 9618 · 2025 May/June · Paper 11 · Q2(b)(i)

2 Programmers in a software development company take part in live video conferences to discuss their work.

The live video conferences take place using real-time bit streaming. The video is compressed before it is transmitted.

(b) **(i)** Explain the reasons why a video is compressed before it is transmitted using real-time bit streaming.

MS 9618 · 2025 May/June · Paper 11 · Q2(b)(i)

Question	Answer	Marks
2(b)(i)	**1 mark** each to **max 4** e.g. • Video is data-intensive • The file size needs reducing in order to • reduce the amount of bandwidth used • and reduce buffering • This means people are not behind in the conversation • and people with lower bandwidth can still take part	**4**

MS **9608** 旧考纲 9608 · 2016 Oct/Nov · Paper 11 · Q6

6 Any **four** from:

• User needs high-speed broadband (connection)
• Data is streamed to a buffer (in the computer)
• Buffering stops video pausing as bits streamed
• As buffer is emptied, it fills up again so that viewing is continuous
• Actual playback is (a few seconds) behind the time the data is received by computer [4]

同类 **MS**（已包含在上图中）：9608 · 2016 Oct/Nov · Paper 13 · Q6

MS 9608 旧考纲 9608 · 2020 Oct/Nov · Paper 11 · Q5(a)(ii)

Question	Answer	Marks
5(a)(ii)	**1 mark** per bullet point to **max 3** • Insufficient bandwidth // slow internet connection • … experiencing problems with buffering • Video is too high quality to stream in real-time • Congestion on the home network • Too much demand for the video from the supplier • Too many applications running on Oscar's computer • Oscar is trying to watch the video in High Definition, his friend is watching the video at a lower resolution	**3**

12 The World Wide Web (WWW) and the internet | WWW 与互联网的区别

12.1 Using the internet and/or the WWW (scenario)

题目情景 **QP** 9618 · 2021 May/June · Paper 11 · Q4(d)

4 Melinda and her friends set up a peer-to-peer network between their computers to share data.

(d) Melinda sends emails from her webmail account (email account accessed through a website).

Explain whether Melinda is using the internet, or the World Wide Web (WWW), or both.

MS 9618 · 2021 May/June · Paper 11 · Q4(d)

Question	Answer	Marks
4(d)	**1 mark** for identifying that she is using both. **1 mark** per bullet point for justification • using internet because sending data on **the infrastructure** • using WWW because accessing a **website** (that is stored on a web server operated by the webmail) that is part of the WWW	**3**

同类 **MS**（已包含在上图中）：9618 · 2021 May/June · Paper 13 · Q4(d)

12.2 Differences between the WWW and the internet

9618 MS 里没有直接问两者区别的题，下面用 9608 旧考纲补充。

MS **9608** 旧考纲 9608 · 2015 May/June · Paper 11 · Q5(b)

(b) **maximum** of **two marks** for Internet references and **maximum** of **two marks** for world wide web references

Internet

- massive network of networks/interconnected network of computer devices
- Internet stands for Interconnected Networks
- uses TCP/IP protocol

World Wide Web (www)

- is a collection of (multimedia) web pages/documents
- ...stored on websites
- http/protocols used to transmit data
- web pages are written in HTML
- URLs specify the location of the web pages
- web documents are accessed using browsers **[3]**

同类 **MS**（已包含在上图中）：9608 · 2015 May/June · Paper 12 · Q5(b)；9608 · 2016 May/June · Paper 13 · Q6(a)；9608 · 2018 May/June · Paper 11 · Q5(b)

+ 额外得分点 **9608** 旧考纲 9608 · 2018 May/June · Paper 11 · Q5(b)

- The World Wide Web is accessed over the Internet

13 Hardware used to support the internet | 支持互联网的硬件

13.1 Modems and dedicated lines

MS 9618 · 2025 May/June · Paper 11 · Q2(c)

Question	Answer	Marks
2(c)	**1 mark** each modems: • Converts digital data into analogue for transmission down phone lines // Converts analogue data into digital after transmission down phone lines dedicated lines: • Used to provide a direct / private connection which therefore provides faster transmission	**2**

13.2 PSTN

MS 9618 · 2024 May/June · Paper 12 · Q3(c)(ii)

Question	Answer	Marks
3(c)(ii)	**1 mark** each to **max 2**: • The PSTN consists of many different types of communication lines • … therefore the digital data may need to be converted into a different form/analogue signal • Data is transmitted in both directions at the same time // duplex data transmission • Using a PSTN the communication passes through different switching centres/ISPs	**2**

同类 MS（已包含在上图中）：9608 · 2019 May/June · Paper 12 · Q1(b)(i)

MS 9608 旧考纲 9608 · 2015 May/June · Paper 11 · Q5(a)

5 (a)

Description	Conventional telephone using PSTN	Internet-based system
connection only in use whilst sound is being transmitted		✓
dedicated channel used between two points for the duration of the call	✓	
connection maintained throughout the telephone call	✓	
encoding schemes and compression technology used		✓
lines remain active even during a power outage	✓	

[5]

同类 MS（已包含在上图中）：9608 · 2015 May/June · Paper 12 · Q5(a)

13.3 Dedicated lines: benefit and drawback

MS 9608 旧考纲 9608 · 2019 May/June · Paper 12 · Q1(b)(ii)

Question	Answer	Marks
1(b)(ii)	**1 mark** for benefit, **1 mark** for drawback Benefit • (Probably) faster connection / communication / transmission of data • (Usually) more consistent transmission speed • Improved security Drawback • Expensive to set-up / maintain • Disruption to the dedicated line would leave no alternative	**2**

13.4 Cell phone network

MS 9618 · 2025 May/June · Paper 12 · Q6(b)

Question	Answer	Marks
6(b)	**1 mark** each to **max 4** e.g. • Land is split into cells designed for maximum line of sight • Each cell has a tower with an antenna which receives and transmits data • Data is transmitted between the tower and the phone • Data transmission is wireless using low power radio signals/frequencies • Multiple devices can communicate simultaneously with the same tower	**4**

13.5 Ways to connect to the internet (dedicated line / cell phone network / satellite)

MS **9608** 旧考纲 9608 · 2020 Oct/Nov · Paper 13 · Q4(b)

Question	Answer	Marks
4(b)	**1 mark** for identification, **1 mark** for further description • Dedicated lines / leased line services • Connection that is only used for that business/organisation // permanent connection • Cell phone network • Send data to cell towers over mobile connection • Satellite • Send data to satellites in orbit	**4**

同类 **MS**（已包含在上图中）：9608 · 2021 Oct/Nov · Paper 12 · Q3(a)

14 IP addresses | IP 地址

14.1 Format of IPv4 and IPv6

MS 9618 · 2025 May/June · Paper 12 · Q6(c)

Question	Answer	Marks
6(c)	**1 mark** for each correct word • **full stops** • **hexadecimal** • **zeros** • **128** • **dynamic** • **private** IPv4 is made of four groups of 8-bit numbers separated by **full stops**. Each IPv4 address is 32 bits. IPv6 is made of eight groups of 4 **hexadecimal** numbers separated by colons. Multiple consecutive groups of **zeros** can be replaced with a double colon. Each IPv6 address is **128** bits. A **dynamic** IP address can change each time the computer connects to a network. A **private** IP address can only be accessed by other devices in the same LAN and is assigned by the router within the LAN.	**6**

同类 **MS**（已包含在上图中）：9618 · 2024 May/June · Paper 11 · Q8(b)(ii)；9618 · 2023 May/June · Paper 11 · Q4(d)(i)；9618 · 2023 Oct/Nov · Paper 12 · Q7(a)

+ 额外得分点 9618 · 2024 May/June · Paper 11 · Q8(b)(ii)

- IPv4 is usually represented in **denary** whilst IPv6 is usually represented in **hexadecimal**
- IPv4 groups are between 0 and 255 whilst IPv6 is between 0 and FFFF

+ 额外得分点 9618 · 2023 May/June · Paper 11 · Q4(d)(i)

An IPv4 address contains **4** groups of digits. Each group is represented in **8** bits and the groups are separated by full stops.

An IPv6 address contains **8** groups of digits. Each group is represented in **16** bits. Multiple groups that only contain zeros can be replaced with a **::** // **double colon**.

14.2 Valid or invalid IP address (application)

题目情景 **QP** 9618 · 2025 Oct/Nov · Paper 11 · Q7(b)

(b) Consider the following IP address:

256.0.0.A

Circle whether this IP address is IPv4, IPv6 or an invalid IP address.

IPv4 IPv6 Invalid

Justify your choice.

MS 9618 · 2025 Oct/Nov · Paper 11 · Q7(b)

Question	Answer	Marks
7(b)	**1 mark** per bullet point for justification, **max 2** marks Choice: Invalid (no mark for the choice) Justification: • It is not IPv4 because the maximum denary value should be 255, this uses 256 • It is not IPv6 because it uses a full stop to separate sections and not colon // It is not IPv6 because there are not enough groups	**2**

14.3 How an IP address is associated with a device

MS 9618 · 2024 May/June · Paper 11 · Q8(b)(i)

Question	Answer	Marks
8(b)(i)	**1 mark** for: to **be visible** to and accessible by other devices on the internet	**1**

MS 9618 · 2024 Oct/Nov · Paper 13 · Q9(d)(i)

Question	Answer	Marks
9(d)(i)	**1 mark** for public IP address and **1 mark** for private IP address: Public IP address: • So that the router is visible to the Internet/external network/WAN Private IP address: • So that the router is identified to computers within the LAN	**2**

9618 MS 只考过 public IP 的用途，下面用 9608 旧考纲补充 IP 地址本身的作用。

MS 9608 旧考纲 9608 · 2018 Oct/Nov · Paper 12 · Q2(b)

Question	Answer	Marks
2(b)	**1 mark** per bullet point to **max 2** • Gives each device on a network an identifier // IP address used to locate a device on a network • Each address is unique within the network • Allows a device/gateway/node to send data to the correct destination / a specific device/gateway/node	**2**

MS 9608 旧考纲 9608 · 2020 Oct/Nov · Paper 11 · Q8(a)

Question	Answer	Marks
8(a)	**1 mark** per bullet point to **max 4** • The router needs a public IP address so it can be identified on the Internet • The router needs a private IP address so it can be identified on the home network • The router has a public and a private IP address so that it can route data between the two networks (home and Internet) • The laptop needs a private IP address so it can be identified on the home network // so the router knows where to send data • The laptop does not have a public IP address because it does not connect directly to the Internet • ... this is more secure because it hides the laptop from the outside world • … all data from the Internet must be transmitted via the router	**4**

同类 **MS**（已包含在上图中）：9608 · 2020 Oct/Nov · Paper 13 · Q4(a)(i)；9608 · 2020 Oct/Nov · Paper 13 · Q4(a)(ii)

14.4 Public / private / static / dynamic IP addresses: definitions

MS 9618 · 2023 Oct/Nov · Paper 12 · Q7(d)

Question	Answer	Marks
7(d)	**1 mark** for each highlighted area **Type of IP address** / **Description** public IP address — an IP address that is assigned to a device to allow direct access on the internet static IP address — an IP address that is fixed / doesn't change each time a device re-joins a LAN / network private IP address — an IP address used for internal LAN communication only dynamic IP address — an IP address that may be refreshed / changed each time a **device re-joins** a LAN / network	**4**

同类 **MS**（已包含在上图中）：9618 · 2025 Oct/Nov · Paper 11 · Q7(a)；9618 · 2024 May/June · Paper 13 · Q5(d)；9618 · 2026 May/June · Paper 11 · Q7(c)(i)

14.5 Public vs private IP: differences and implications for security

9618 MS 里没有专门比较 public 与 private IP 的题，下面用 9608 旧考纲补充。

MS **9608** 旧考纲 9608 · 2016 May/June · Paper 11 · Q7(c)

(c) **Two** points from: **[2]**

- Public address can be reached across the Internet.
- Private address can only be reached internally/through the LAN/Intranet // private address cannot be reached across the Internet.
- NAT (Network Address Translation) is necessary for a private IP address to access the Internet directly.
- A private address is more secure than a public address // A public address is less secure than a private address.
- Public addresses are provided by ISP / assigned by InterNIC // Private addresses are assigned by the router (of the network concerned).
- Public addresses are unique (to the Internet) // Private addresses (are unique within their network, but) can be duplicated within other (discrete) networks.
- 10.0.0.1 to 10.255.255.254 and 172.16.0.1 to 172.31.255.254 and 192.168.0.1.to 192.168.255.254 form the private address space // IP addresses from the private address space are never assigned as public.

同类 **MS**（已包含在上图中）：9608 · 2016 May/June · Paper 12 · Q7(c)；9608 · 2021 May/June · Paper 11 · Q9(a)；9608 · 2021 Oct/Nov · Paper 13 · Q3(a)；9608 · 2018 Oct/Nov · Paper 13 · Q4(c)；9608 · 2018 Oct/Nov · Paper 11 · Q2(b)(ii)

+ 额外得分点 **9608** 旧考纲 9608 · 2021 Oct/Nov · Paper 13 · Q3(a)

- Public is used to get internet service

+ 额外得分点 **9608** 旧考纲 9608 · 2021 Oct/Nov · Paper 13 · Q3(a)

- Private is used to communicate within a network

MS **9608** 旧考纲 9608 · 2020 May/June · Paper 13 · Q1(c)

Question	Answer	Marks
1(c)	**1 mark** per bullet point • Improved security because the IP address is not visible outside the network • An internet presence is not required for each employee computer • Only the router needs a public IP address, as only the router needs to be externally visible • Reduces number of (public) IP addresses needed	**2**

14.6 Static vs dynamic IP: why a server uses a static IP

MS **9608** 旧考纲 9608 · 2020 Oct/Nov · Paper 12 · Q7(b)

Question	Answer	Marks
7(b)	**1 mark** per bullet point to **max 3** • Static IP does not change whereas a dynamic IP address does change • … the DNS does not need updating • … which might be delayed causing 'address not found' errors • The webserver may be accessed directly using just the IP address // the IP address is still held in cache memory	**3**

14.7 Subnetting: purpose

MS 9618 · 2024 Oct/Nov · Paper 13 · Q9(d)(ii)

Question	Answer	Marks
9(d)(ii)	**1 mark** for each bullet point (**max 2**) • It allows the network to be divided into smaller networks • … which reduces traffic in some parts of the network // reduces congestion • … because traffic only travels through the parts necessary • … which hides the complexity of network • … and allows for easier maintenance of the network	**2**

14.8 Subnetting: benefits / why it is used

MS 9618 · 2023 Oct/Nov · Paper 12 · Q7(b)(iii)

Question	Answer	Marks
7(b)(iii)	**1 mark** for each use (**max 2**) and **1 mark** for corresponding expansion (**max 2**) • To improve the security of the LAN • … so that devices do not receive unintended data • … so that a compromised device does not expose the whole network • ... so not all devices can access all segments • To make the network management easier • … because faults can be isolated more efficiently • ... by appropriate example • To make the network easier to expand // For better control of network growth • … by allowing for greater range of IP addresses to be available • To improve network performance • To reduce network congestion • … by localising network communications // by dividing data between segments • … so that devices are not flooded with data • … because data sent between devices on the same subnet stays within the subnet	**4**

同类 **MS**（已包含在上图中）：9618 · 2023 May/June · Paper 11 · Q4(d)(ii)；9618 · 2026 May/June · Paper 11 · Q7(c)(ii)；9618 · 2022 Oct/Nov · Paper 13 · Q7(c)(i)

+ 额外得分点 9618 · 2026 May/June · Paper 11 · Q7(c)(ii)

- Congestion is reduced // fewer collisions happen

+ 额外得分点 9618 · 2023 May/June · Paper 11 · Q4(d)(ii)

- Allows for easier maintenance
- … because only one subnetwork may need taking down/changing while the rest of the network can continue

14.9 Subnetting: network ID and host ID

MS 9618 · 2023 May/June · Paper 13 · Q2(e)

Question	Answer	Marks
2(e)	**1 mark** for identification • IP address is made up of a network ID and a host ID **1 mark** each to **max 2** for description • Each device in a subnetwork has the same network ID // Each subnetwork has a different network ID • Every device **in each subnetwork** has a different host ID but the same network ID // the host ID uniquely identifies the device **within the subnetwork**	**3**

題目情景 QP 9618 · 2022 Oct/Nov · Paper 13 · Q7(c)(ii)

(ii) A subnet mask is used when subnetting a LAN.

Two devices on the LAN are located in different subnetworks.

The IP addresses and corresponding subnet masks are shown:

Device IP address	Subnet mask
`10.10.12.1`	`255.0.0.0`
`192.168.12.4`	`255.255.255.0`

Identify the following network ID and host ID.

The network ID for the device with the IP address `10.10.12.1`

.....

The host ID for the device with the IP address `192.168.12.4`

MS 9618 · 2022 Oct/Nov · Paper 13 · Q7(c)(ii)

Question	Answer	Marks
7(c)(ii)	**1 mark** for each correct answer: • network ID = 10 • host ID = 4	**2**

15 URL and DNS | URL 与 DNS

15.1 How a web browser uses a URL to access a web page

MS 9618 · 2025 Oct/Nov · Paper 11 · Q7(c)

Question	Answer	Marks
7(c)	**1 mark** per bullet point, **max 4** marks • The browser checks its cache for the URL • The browser parses the URL and splits it into its component parts • The browser finds the IP address for the domain name by querying a DNS server • The browser receives the matching IP address from the DNS server • The browser creates connection with the web server with the matching IP address • A request for the resource is sent to the web server with that IP address • The browser renders and displays the result • The IP address is stored in the browser cache for future use	**4**

15.2 What a URL is and the role of DNS

9618 MS 里没有专门定义 URL、讲 DNS 解析过程的题，下面用 9608 旧考纲补充。

MS **9608 旧考纲** 9608 · 2015 Oct/Nov · Paper 12 · Q9(b)

(b) Maximum **3 marks** for URL and maximum **3 marks** for DNS

– URL = uniform resource locator
– reference address to a resource/website on the Internet
– includes protocol used /includes domain name

– browser software sends URL to DNS

– DNS = Domain name system // Domain name service // Domain name server
– the true domain name (e.g. cie_exams.co.uk) is resolved/turned into an IP address by DNS
– DNS server stores a database/list of URLs with matching IP address
– a DNS server may need to pass a request onto another DNS server, (if URL not in its database)
– DNS server adds returned IP address and URL to its cache/database
– DNS server may return an error message if requested domain name is invalid or does not exist
– the DNS service has a hierarchy/or by example
– DNS server returns IP address to browser [4]

同类 **MS**（已包含在上图中）：9608 · 2016 May/June · Paper 13 · Q6(e)；9608 · 2017 May/June · Paper 11 · Q7(b)；9608 · 2019 May/June · Paper 11 · Q1(b)；9608 · 2020 Oct/Nov · Paper 11 · Q8(b)；9608 · 2021 Oct/Nov · Paper 13 · Q3(b)

15.3 Domain name and IP address

MS 节选 **9608 旧考纲** 9608 · 2021 May/June · Paper 13 · Q6(b)

- A domain name is a memorable form for an IP address

MS 节选 **9608 旧考纲** 9608 · 2021 May/June · Paper 13 · Q6(b)

- Each domain name is linked to an IP address

附录 MS 来源索引（按场次）

列出本笔记用到的每份 MS 及对应知识点编号，方便回查原卷。

场次 · 卷号	题号 → 知识点（用途）
9618 · 2021 May/June · Paper 11	Q4(a) → 3.4（主图）；Q4(b) → 3.6（主图）；Q4(c)(ii) → 7.7（主图）；Q4(c)(i) → 9.1（同类）；Q4(d) → 12.1（主图）
9618 · 2021 May/June · Paper 12	Q5(c)(i) → 6.5（主图）；Q5(c)(ii) → 6.6（同类）
9618 · 2021 May/June · Paper 13	Q4(a) → 3.4（同类）；Q4(b) → 3.6（同类）；Q4(c)(ii) → 7.7（同类）；Q4(d) → 12.1（同类）
9618 · 2021 Oct/Nov · Paper 11	Q8(a) → 2.4（主图）；Q8(c) → 8.6（主图）
9618 · 2021 Oct/Nov · Paper 12	Q3(b)(i) → 5.5（主图）；Q3(b)(ii) → 9.1（同类）
9618 · 2021 Oct/Nov · Paper 13	Q8(a) → 2.4（同类）；Q8(c) → 8.6（同类）
9618 · 2022 May/June · Paper 12	Q9(b) → 7.3（主图）；Q9(a) → 8.1（主图）；Q9(c) → 10.2（同类）
9618 · 2022 May/June · Paper 13	Q8(a) → 6.1（主图）；Q8(b) → 6.2（主图）；Q8(c) → 6.5（同类）；Q8(c) → 6.5（额外得分点）；Q8(c) → 6.6（同类）
9618 · 2022 Oct/Nov · Paper 11	Q8 → 10.2（主图）
9618 · 2022 Oct/Nov · Paper 12	Q10(a) → 5.5（同类）；Q10(b)(ii) → 9.1（主图）
9618 · 2022 Oct/Nov · Paper 13	Q7(a) → 6.2（同类）；Q7(b) → 7.6（主图）；Q7(c)(i) → 14.8（同类）；Q7(c)(ii) → 14.9（主图）
9618 · 2023 May/June · Paper 11	Q4(d)(i) → 14.1（同类）；Q4(d)(i) → 14.1（额外得分点）；Q4(d)(ii) → 14.8（同类）；Q4(d)(ii) → 14.8（额外得分点）
9618 · 2023 May/June · Paper 12	Q1(a) → 1.1（主图）；Q1(b) → 2.1（同类）；Q1(e) → 4.2（主图）；Q1(c) → 5.5（主图）；Q1(d) → 10.1（主图）
9618 · 2023 May/June · Paper 13	Q2(a) → 2.3（主图）；Q2(c) → 4.1（额外得分点）；Q2(b)(i) → 5.8（主图）；Q2(b)(ii) → 5.9（主图）；Q2(d) → 7.9（主图）；Q2(e) → 14.9（主图）
9618 · 2023 Oct/Nov · Paper 11	Q2(d) → 5.7（主图）；Q2(c) → 6.6（同类）；Q2(b) → 8.2（主图）；Q2(a) → 9.1（同类）；Q2(a) → 9.1（额外得分点）
9618 · 2023 Oct/Nov · Paper 12	Q7(b)(i) → 5.5（同类）；Q7(b)(ii) → 5.6（主图）；Q7(c) → 10.2（同类）；Q7(a) → 14.1（同类）；Q7(d) → 14.4（主图）；Q7(b)(iii) → 14.8（主图）
9618 · 2023 Oct/Nov · Paper 13	Q3(a)(i) → 6.3（主图）；Q3(a)(ii) → 6.7（主图）
9618 · 2024 May/June · Paper 11	Q5(a) → 3.1（额外得分点）；Q8(a) → 5.1（主图）；Q2(e)(i) → 11.1（主图）；Q2(e)(ii) → 11.3（主图）；Q8(b)(ii) → 14.1（同类）；Q8(b)(ii) → 14.1（额外得分点）；Q8(b)(i) → 14.3（主图）
9618 · 2024 May/June · Paper 12	Q3(b) → 4.1（同类）；Q3(c)(i) → 9.1（同类）；Q3(c)(ii) → 13.2（主图）
9618 · 2024 May/June · Paper 13	Q5(b) → 5.6（同类）；Q5(a)(i) → 6.2（同类）；Q5(a)(ii) → 6.4（主图）；Q5(c)(i) → 10.2（同类）；Q5(c)(ii) → 10.3（主图）；Q5(d) → 14.4（同类）
9618 · 2024 Oct/Nov · Paper 13	Q9(a) → 2.2（主图）；Q9(b) → 7.1（主图）；Q9(c) → 11.4（同类）；Q9(c) → 11.4（额外得分点）；Q9(d)(i) → 14.3（主图）；Q9(d)(ii) → 14.7（主图）
9618 · 2025 May/June · Paper 11	Q2(a) → 11.4（主图）；Q2(b)(i) → 11.7（主图）；Q2(c) → 13.1（主图）
9618 · 2025 May/June · Paper 12	Q6(a) → 2.3（额外得分点）；Q6(e) → 8.3（主图）；Q6(d) → 8.8（主图）；Q6(b) → 13.4（主图）；Q6(c) → 14.1（主图）
9618 · 2025 May/June · Paper 13	Q3(c) → 3.1（主图）
9618 · 2025 Oct/Nov · Paper 11	Q7(b) → 14.2（主图）；Q7(a) → 14.4（同类）；Q7(c) → 15.1（主图）
9618 · 2025 Oct/Nov · Paper 12	Q5(a) → 2.1（主图）；Q5(b) → 5.5（同类）；Q5(c) → 5.6（同类）；Q8(a)(i) → 6.5（同类）；Q8(a)(ii) → 6.6（主图）；Q5(d) → 10.2（同类）；Q5(d) → 10.2（额外得分点）
9618 · 2025 Oct/Nov · Paper 13	Q4(a) → 7.7（同类）；Q4(a) → 7.7（额外得分点）；Q4(b) → 7.8（主图）
9618 · 2026 May/June · Paper 11	Q7(a) → 4.1（主图）；Q7(b)(i) → 5.2（主图）；Q7(b)(ii) → 5.3（主图）；Q7(b)(i)（见 → 10.1（同类）；Q7(c)(i) → 14.4（同类）；Q7(c)(ii) → 14.8（同类）；Q7(c)(ii) → 14.8（额外得分点）
9618 · 2026 May/June · Paper 12	Q6(d) → 6.5（额外得分点）；Q6(d) → 6.6（额外得分点）
9618 · 2026 May/June · Paper 13	Q3(c)(ii) → 6.2（同类）；Q3(c)(iii) → 6.5（同类）
9608 · 2015 May/June · Paper 11	Q5(b) → 12.2（主图）；Q5(a) → 13.2（主图）
9608 · 2015 May/June · Paper 12	Q5(b) → 12.2（同类）；Q5(a) → 13.2（同类）
9608 · 2015 Oct/Nov · Paper 11	Q7(a)(i) → 3.1（主图）；Q7(a)(ii) → 3.2（主图）；Q6 → 7.4（主图）；Q1(a) → 11.1（主图）；Q1(c) → 11.5（主图）；Q1(b)(i) → 11.6（主图）；Q1(b)(ii) → 11.6（主图）
9608 · 2015 Oct/Nov · Paper 12	Q9(b) → 15.2（主图）
9608 · 2015 Oct/Nov · Paper 13	Q7(a)(ii) → 3.2（同类）；Q6 → 7.4（同类）；Q1(a) → 11.1（同类）；Q1(c) → 11.5（同类）
9608 · 2016 May/June · Paper 11	Q7(c) → 14.5（主图）
9608 · 2016 May/June · Paper 12	Q7(c) → 14.5（同类）
9608 · 2016 May/June · Paper 13	Q6(c) → 11.2（主图）；Q6(a) → 12.2（同类）；Q6(e) → 15.2（同类）
9608 · 2016 Oct/Nov · Paper 11	Q6 → 11.7（主图）
9608 · 2016 Oct/Nov · Paper 13	Q6 → 11.7（同类）
9608 · 2017 May/June · Paper 11	Q7(b) → 15.2（同类）
9608 · 2018 May/June · Paper 11	Q5(a) → 8.4（同类）；Q5(b) → 12.2（同类）；Q5(b) → 12.2（额外得分点）
9608 · 2018 Oct/Nov · Paper 11	Q2(b)(ii) → 14.5（同类）

9608 · 2018 Oct/Nov · Paper 12	Q2(c)(i) → 7.4（同类）；Q2(c)(ii) → 7.5（主图）；Q2(b) → 14.3（主图）
9608 · 2018 Oct/Nov · Paper 13	Q7(a) → 7.7（同类）；Q7(b) → 7.8（同类）；Q4(c) → 14.5（同类）
9608 · 2019 May/June · Paper 11	Q1(b) → 15.2（同类）
9608 · 2019 May/June · Paper 12	Q1(d) → 8.4（主图）；Q1(b)(i) → 13.2（同类）；Q1(b)(ii) → 13.3（主图）
9608 · 2019 Oct/Nov · Paper 11	Q4(a)(i) → 3.1（同类）；Q4(c)(i) → 7.4（同类）；Q4(c)(ii) → 7.5（额外得分点）
9608 · 2019 Oct/Nov · Paper 12	Q6(e)(i) → 11.4（同类）
9608 · 2020 May/June · Paper 13	Q1(b) → 3.1（同类）；Q1(a) → 7.2（主图）；Q1(c) → 14.5（主图）
9608 · 2020 Oct/Nov · Paper 11	Q5(a)(ii) → 11.7（主图）；Q8(a) → 14.3（主图）；Q8(b) → 15.2（同类）
9608 · 2020 Oct/Nov · Paper 12	Q7(b) → 14.6（主图）
9608 · 2020 Oct/Nov · Paper 13	Q4(b) → 13.5（主图）；Q4(a)(i) → 14.3（同类）；Q4(a)(ii) → 14.3（同类）
9608 · 2021 May/June · Paper 11	Q9(b) → 8.4（主图）；Q9(a) → 14.5（同类）
9608 · 2021 May/June · Paper 13	Q6(b) → 15.3（节选）
9608 · 2021 Oct/Nov · Paper 11	Q11(a) → 3.1（同类）
9608 · 2021 Oct/Nov · Paper 12	Q3(a) → 13.5（同类）
9608 · 2021 Oct/Nov · Paper 13	Q3(a) → 14.5（同类）；Q3(a) → 14.5（额外得分点）；Q3(b) → 15.2（同类）